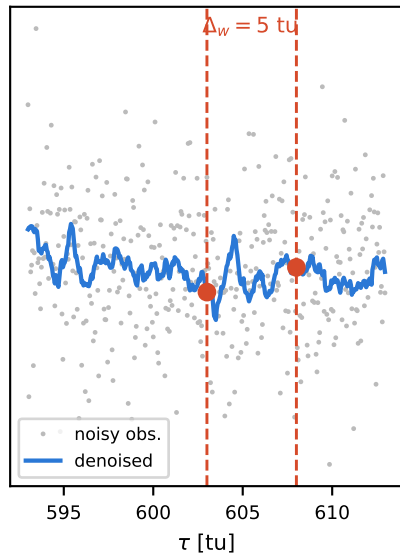
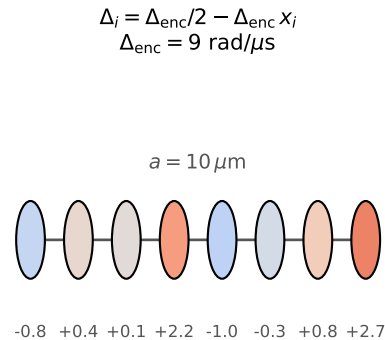


(a) Denoised input window
 $\mathbf{w} \in \mathbb{R}^8$: 4 states $\times$ 2 samples



(b) Local-detuning encoding
 8 atoms, one feature per site



(c) Neutral-atom evolution
 independent run per probe time

$$H = \sum_i \left[\frac{\Omega}{2} \sigma_x^{(i)} - \Delta_i n_i \right] + \sum_{i < j} V_{ij} n_i n_j$$

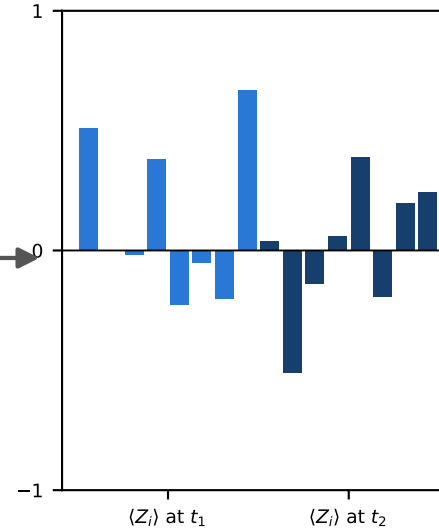
$$V = C_6/a^6 = 5.42 \text{ rad}/\mu\text{s}$$

$\Omega = 2\pi \text{ rad}/\mu\text{s}$ (quench)

projective measurement

t_1 t_2

(d) Z / ZZ embedding
 $E \in \mathbb{R}^{N_a N_t}$ (500 shots)



(e) Ridge readout
 finite-horizon increment prediction

$$\hat{\mathbf{y}} = \mathbf{W}_{\text{ridge}} E$$

RidgeCV, fitted on training anchors only

target: $\mathbf{y} = \mathbf{x}_{\text{clean}}(\tau + H) - \mathbf{x}_{\text{clean}}(\tau)$
 $H \in \{1, 2, 3\} \tau_c$